Natural Language Processing

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The National Higher School of AI

Chapter 0

Course Organisation

Course Objectives

- Introduction to the fundamental NLP concepts.
- Understanding of the basic text processing steps: Lexical, morphological, syntactic and semantic analyses.
- Development of in-depth knowledge of the fundamental algorithms (logical or statistical) for processing linguistic information.
- Neural Nets & Transformer architectures for NLP
- The power of Machine Learning techniques for NLP tasks and applications: word sense disambiguation, Question-Answering systems, Machine Translation, NL Generation, etc.

Course instructors

- Lectures: **Prof. Ahmed Guessoum**

  Sun.: 8:30-10:10 (Sec. 2 then Sec. 1; Amphi. 4)

  Mon.: 8:30-10:10 (Sec. 1 then Sec. 2; Amphi. 7)

  Office Hours (Room L2-102):

  Sun. 11:45 – 13:00

  Mon. 11:45 – 13:00

- Labs :

Dr. Mohamed Brahimi G6: Mon. 15:10 – 18:20, Lab 12 G10: Wed. 15:10 – 18:20, Lab 7	**Dr. Zineb Djouamai** G5: Mon. 15:10 – 18:20, Lab 1
Dr. Elhocine Boutellaa G2: Sun. 13:30 – 16:40, Lab 2 G4: Tue. 10:10 – 13:20, Lab 8 G7: Mon. 13:30 – 16:40, Lab 8	**Dr. Mohamed S. Hadj Ameur** G1: Sun. 13:30 – 16:40, Lab 3 G8: Mon. 13:30 – 16:40, Lab 6 G3: Tue. 08:30 – 11:40, Lab 4 G9: Tue. 13:30 – 16:40, Lab 11

Course Organisation

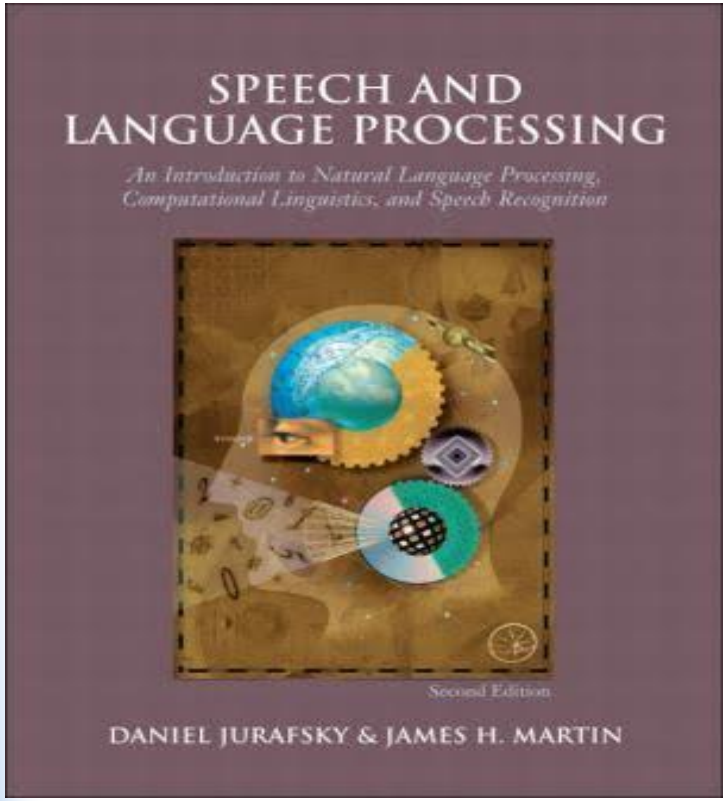
- **Prerequisites**:
  - Good programming skills
  - Some fundamentals of Machine Learning
- **Grading**:
  - Continuous assessment:
    - Instructors' Assessment (Labs) /10
    - 1 Project /25
    - 1 Midterm Exam (ME) /25

    Counts for 40%
  - Final exam (FE): /20

    Counts for 60%
- Course average = (2*(A + P + ME)/3 + 3*FE) / 5

Textbook

Jurafsky, D. and Martin, J.H. (2024) *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models* (3rd ed. draft), Prentice Hall.
https://web.stanford.edu/~jurafsky/slp3/ed3bookaug20_2024.pdf



The draft version of the 3rd edition will be used; it has been updated to include more recent topics such as Vector Semantics and Embeddings, ANNs and Deep Learning for NLP, Transformers, etc.

This course **slides** will largely be based on this textbook and its authors' slides with light customisations whenever seen fit.

Other references

- Delip Rao and Brian McMahan (2019) *Natural Language Processing with PyTorch*, O'Reilly.
- Lewis Tunstall, Leandro von Werra, and Thomas Wolf (2022) *Natural Language Processing with Transformers,* O'Reilly Media.
- Chris Manning and Hinrich Schutze, "Foundations of Statistical Natural Language Processing", The MIT Press, 1999
- Gerald Gazdar & Chris Mellish (1989) "Natural Language Processing in Prolog", Addison-Wesley.
- Jacob Eisenstein (2019) *Natural Language Processing*, The MIT Press.
- Frederick Jelinek (1998) "Statistical Methods for Speech Recognition", The MIT Press.

Commitment!

- Lectures:
  - Attendance will help you enormously in understanding the material and in better answering exam questions
  - I will NOT take the attendance! You are mature enough to decide if you need to attend or not.
  - No late students' arrivals will be accepted, i.e. no one will be accepted, after the official start time.
- Lab sessions:
  - Attendance is mandatory thus the Instructors' mark will depend on it as well as your work and participation in the labs.
  - The exclusion rules will be applied.
- The Mini project rules and late submission policy will be detailed in the project statement in due time.

Assessments

- **Midterm exams period**: 14–21/11/2026
  (NLP Exam to be confirmed by the administration)
- Absence at exam?
  - Bring an official justification validated by the administration
  - Make-up exam?
- **Mini-Project**: Work in teams.
  - Release date: 10/10/2026 at the latest in shaa Allaah;
  - Due date: 02/01/2027 (No deadline postponement!)
- No cheating (plagiarism or other type of misconduct) will be tolerated. ➔ Very serious consequences.

Course Content (cont.)

Part 1: NLP Fundamentals

- Introduction and Overview
- Language and Text Processing
- N-gram Language Models
- Vector Semantics and Embeddings
- Neural Networks and Neural Language Models
- Sequence Labelling for Parts of Speech and Named Entities
- Transformers and Pre-trained Language Models

Course Content (cont.)

Part 2: Annotating Language Structure

- Constituency Parsing

  The following topics If time permits
- Dependency Parsing
- Logical Representations of Sentence Meaning
- Computational Semantics and Semantic Parsing

Part 3: NLP Applications (Project-based learning)

- Question Answering
- Chatbots
- Machine Translation

Part 4: Generative AI for NLP (Project-based learning)

Course Teaching

- **Lecture PowerPoints:**
  - All slides are based on Jurafsky and Martin's textbook
  - They will be sent to you as we cover different chapters/sections.

- **Labs**:
  - Use of Jupyter/Python

Questions?